BUSINESS DETECTIVE

Customer Support Investigation Report

1. Executive Summary

CSAT moved 3.7 -> 3.56 while monthly ticket volume changed +35.4% and SLA breach rate moved 5.2% -> 11.2%. The largest worsening was in: IssueType = Technical: SLA breach 13.7% -> 32.1%, avg resolution 12.82h -> 18.46h, CSAT 3.33 -> 2.96.

2. Investigation Question

Why are customer support issues increasing, and where are the operational problems occurring?

3. Observed Symptoms

Comparison window: 2026-01..2026-06 (before) vs 2026-07..2026-09 (after). Average resolution 7.71h -> 9.72h.

4. KPI Overview

tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct	avg_first_response_min
8000.0	7.6	8.52	3.64	6.1	5.0	73.0

5. SLA Analysis

segment	tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct
Critical	665	40.6	8.74	3.65	5.6	9.2
High	1744	11.3	8.45	3.64	5.5	5.0
Low	1910	0.2	8.43	3.66	6.0	4.0
Medium	3681	3.8	8.56	3.64	6.4	4.6

6. Resolution-Time Analysis

segment	tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct
Account	1567	1.4	5.52	3.86	4.7	3.4
Delivery	1618	3.3	7.32	3.7	5.0	4.2
Login	2008	0.4	3.7	3.98	4.9	4.3
Payment	1187	15.4	13.29	3.35	6.1	4.8
Technical	1620	21.1	15.09	3.18	9.8	8.1

7. Issue-Type Analysis (largest SLA worsening)

dimension	segment	breach_before	breach_after	res_before	res_after	csat_before	csat_after
IssueType	Technical	13.7	32.1	12.82	18.46	3.33	2.96
IssueType	Payment	9.7	23.9	10.94	16.8	3.5	3.14
Priority	High	6.0	18.9	7.51	9.79	3.72	3.51
Priority	Critical	36.7	46.3	7.8	10.12	3.68	3.61
Region	South	7.0	14.8	8.6	11.43	3.64	3.5
Product	Hardware	5.0	12.5	7.86	9.55	3.69	3.59
CustomerSegment	Enterprise	5.1	11.9	7.59	10.07	3.69	3.53
Channel	Chat	4.5	11.2	7.39	9.47	3.81	3.67

8. Channel Analysis

segment	tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct
Chat	2773	7.2	8.24	3.75	6.4	4.7
Email	3580	8.4	9.0	3.49	6.0	5.0
Phone	1647	6.6	7.95	3.78	5.6	5.3

9. Customer Satisfaction

segment	tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct
East	2003	6.9	8.1	3.67	6.2	4.8
North	2376	7.1	8.16	3.66	6.0	5.3
South	1970	10.2	9.76	3.58	5.7	5.2
West	1651	6.2	8.06	3.65	6.4	4.3

Pearson r (resolution hours vs CSAT) = -0.50; (first response vs CSAT) = -0.20. Longer resolution is associated with lower CSAT. Correlation does not establish causation.

10. Reopened / Escalated Ticket Analysis

segment	tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct
No	7515	7.3	8.41	3.7	0.0	4.7
Yes	485	12.6	10.23	2.72	100.0	9.3

segment	tickets	sla_breach_pct	avg_resolution_hrs	avg_csat	reopen_pct	escalation_pct
No	7604	6.9	8.25	3.68	5.8	0.0
Yes	396	22.2	13.7	2.89	11.4	100.0

11. SQL Evidence

Queries used (see the sql/ folder): customer_experience: Satisfaction by channel; customer_experience: CSAT by resolution-time band; customer_experience: Lowest-satisfaction regions; investigation: SLA breach by issue type; investigation: Before vs after by issue type (CTE); investigation: Reopened tickets vs satisfaction; operations: Ticket volume by month; operations:

Escalations by issue type and priority; operations: Products generating most tickets; sla_analysis: Average resolution by priority; sla_analysis: Monthly breach rate with change (window function); sla_analysis: Breach rate by region and priority

12. Root-Cause Candidates

Where SLA performance worsened most
Observed: IssueType = Technical: SLA breach 13.7% -> 32.1%, avg resolution 12.82h -> 18.46h, CSAT 3.33 -> 2.96.
Interpretation: This segment shows the largest worsening and is the strongest lead for the overall decline.
Hypothesis: A capacity, complexity or process change affecting Technical tickets after 2026-07.

Reopened tickets and satisfaction
Observed: Reopened tickets average CSAT 2.72 vs 3.7 for others.
Interpretation: Reopened tickets are associated with poorer customer experience in this dataset.
Hypothesis: Incomplete first-contact resolution may be contributing to lower satisfaction.

Resolution time and satisfaction
Observed: Pearson r (resolution hours vs CSAT) = -0.50; (first response vs CSAT) = -0.20.
Interpretation: Longer resolution is associated with lower CSAT. Correlation does not establish causation.
Hypothesis: Waiting time itself may reduce satisfaction, or hard issues drive both long times and low CSAT.

13. Alternative Explanations

- Ticket mix changed (more complex issues), not agent or process performance.
- Staffing or scheduling changes reduced capacity during the period.
- A product release or policy change created new problem types.
- Survey response behaviour changed, so CSAT shifted without service changing.

14. Recommended Next Tests

- Check staffing, ticket complexity, releases and policy changes touching Technical.
- Examine first-response time, issue complexity and agent/team assignment for reopened tickets.
- Compare CSAT at equal issue type and priority across resolution-time bands.

15. Limitations

Fictional, generated data; observational analysis only; no causal claims; agent comparisons ignore case complexity.

16. Methodology

DuckDB SQL over a single ticket table. Periods: last three months vs earlier months. Segments under 100 tickets excluded from leads. SLA thresholds: Critical 8h, High 16h, Medium 24h, Low 48h.